

# Primary UD, EPR 5 kV - 46 kV

Aluminum Conductor, EPR Insulated
Concentric Neutral, LLDPE Jacket

## Features

Low tension stripping compounds

True Triple extrusion system and closed handling raw materials system, to eliminate any contact with ambient, until extrusion of insulation and shields

RUS approved

## Application

Underground primary residential and commercial distribution circuits

May be used in wet or dry locations, installed in underground ducts or direct burial

## Standards

**ICEA S-94-649**:
Standard for concentric neutral rated cables 5 to 46 kV

**AEIC CS8:**
Specifications for extruded dielectric, shielded power cable, rated 5 to 46 kV

## SPECIFICATIONS

**Maximum operating voltage:**

5 kV - 35 kV, 100% & 133% IL

**Maximum conductor operation temperatures:**

Wet and dry locations:

- Normal: 105°C
- Emergency: 140°C
- Short Circuit: 250°C

## Engineering Information

**Conductor:** Aluminum alloy 1350-H19, compres**sed** Class B stranding, or unilay-compressed per ASTM B 231, or combination unilay (19 wires) per ASTM B 786. *On Request: Copper conductor and/or strand filled*

**Sizes:** 8 AWG - 2 000 kcmil. *On request larger sizes*

**Conductor Shield:** Semi conducting cross-linked polyethylene (XLPE). *On request, super-clean and super-smooth materials*

**Insulation:** Thermoset ethylene propylene rubber (EPR). *On request: Amorphous EPR*

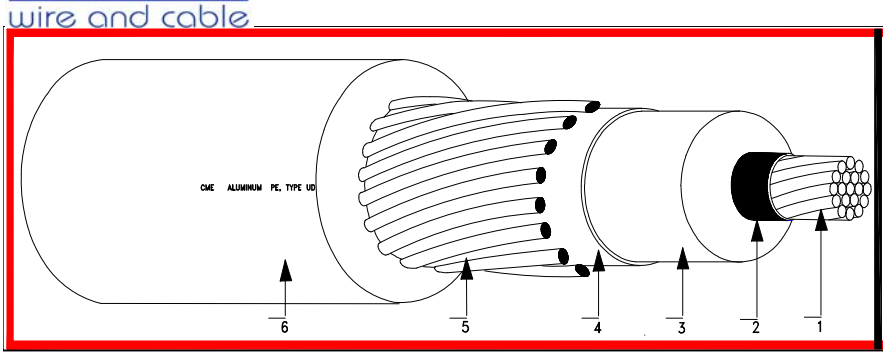
**Insulation Shield:** Semi conducting cross-linked polyethylene (XLPE)

**Concentric Neutral**: Soft-annealed solid copper wires per ASTM B 3, helically applied and uniformly spaced. Full or 1/3 Neutral. *On request: per customer specs.*

**Jacket:** Extruded to fill (Encapsulated) Black sunlight resistant linear low density polyethylene (LLDPE), with three Extruded Red Stripes. *On request: Overlying (Sleeve) Black polyvinyl chloride (PVC), or semiconducting LDPE*

**Configuration Options***: On Request, Triplex or Paralleled configurations*

# *Primary UD, TR-XLPE, 5 kV - 46 kV Aluminum Conductor, TR-XLPE Insulated, Concentric Neutral, LDPE Jacket*




1.– ALUMINUM CONDUCTOR COMPRESSED STRANDED
2.– EXTRUDED CONDUCTOR SHIELD
3.– TR–XLPE INSULATION
4.– EXTRUDED INSULATION SHIELD
5.– CONCENTRIC NEUTRAL
6.– ENCAPSULATED POLYETHYLENE JACKET

*1/3 NEUTRAL* — *15 kV 100% INSULATION LEVEL*

| Size AWG or kcmil | Number of Strands | Insulation Thickness | 1/3 Neutral No. of wires | 1/3 Neutral Size AWG | Jacket Thickness | Approximate Outside Diameter | Approximate Net Weight |
|---|---|---|---|---|---|---|---|
| | | Mils | Mils | Mils | Mils | Inches | Lbs/Mft |
| 2 | 7 | 175 | 6 | 14 | 50 | 1.00 | 435 |
| 1 | 19 | 175 | 6 | 14 | 50 | 1.03 | 470 |
| 1/0 | 19 | 175 | 6 | 14 | 50 | 1.07 | 510 |
| 2/0 | 19 | 175 | 7 | 14 | 50 | 1.12 | 570 |
| 3/0 | 19 | 175 | 9 | 14 | 50 | 1.17 | 652 |
| 4/0 | 19 | 175 | 11 | 14 | 50 | 1.22 | 746 |
| 250 | 37 | 175 | 13 | 14 | 50 | 1.28 | 833 |
| 350 | 37 | 175 | 18 | 14 | 50 | 1.41 | 1064 |
| 500 | 37 | 175 | 16 | 12 | 50 | 1.57 | 1392 |
| 750 | 61 | 175 | 15 | 10 | 80 | 1.88 | 2023 |
| 1000 | 61 | 175 | 16 | 9 | 80 | 2.09 | 2591 |

*FULL NEUTRAL* — *15 kV 100% INSULATION LEVEL*

| Size AWG or kcmil | Number of Strands | Insulation Thickness | Full Neutral No. of wires | Full Neutral Size AWG | Jacket Thickness | Approximate Outside Diameter | Approximate Net Weight |
|---|---|---|---|---|---|---|---|
| | | Mils | Mils | Mils | Mils | Inches | Lbs/Mft |
| 2 | 7 | 175 | 10 | 14 | 50 | 1.00 | 485 |
| 1 | 19 | 175 | 13 | 14 | 50 | 1.03 | 555 |
| 1/0 | 19 | 175 | 16 | 14 | 50 | 1.07 | 631 |
| 2/0 | 19 | 175 | 13 | 12 | 50 | 1.15 | 759 |
| 3/0 | 19 | 175 | 16 | 12 | 50 | 1.20 | 875 |
| 4/0 | 19 | 175 | 13 | 10 | 50 | 1.30 | 1070 |
| 250 | 37 | 175 | 16 | 10 | 50 | 1.36 | 1227 |
| 350 | 37 | 175 | 16 | 9 | 50 | 1.51 | 1554 |

The above data are approximate and subject to normal manufacturing tolerances.

| *1/3 NEUTRAL* | | | | | | | *15 kV 133% INSULATION LEVEL* |
|---|---|---|---|---|---|---|---|
| Size AWG or kcmil | Number of Strands | Insulation Thickness | 1/3 Neutral No. of wires | 1/3 Neutral Size AWG | Jacket Thickness | Approximate Outside Diameter | Approximate Net Weight |
| | | Mils | Mils | Mils | Mils | Inches | Lbs/Mft |
| 2 | 7 | 220 | 6 | 14 | 50 | 1.09 | 499 |
| 1 | 19 | 220 | 6 | 14 | 50 | 1.13 | 535 |
| 1/0 | 19 | 220 | 6 | 14 | 50 | 1.17 | 577 |
| 2/0 | 19 | 220 | 7 | 14 | 50 | 1.21 | 640 |
| 3/0 | 19 | 220 | 9 | 14 | 50 | 1.26 | 724 |
| 4/0 | 19 | 220 | 11 | 14 | 50 | 1.32 | 821 |
| 250 | 37 | 220 | 13 | 14 | 50 | 1.40 | 937 |
| 350 | 37 | 220 | 18 | 14 | 50 | 1.50 | 1150 |
| 500 | 37 | 220 | 16 | 12 | 80 | 1.73 | 1558 |
| 750 | 61 | 220 | 15 | 10 | 80 | 1.98 | 2138 |
| 1000 | 61 | 220 | 16 | 9 | 80 | 2.18 | 2719 |
| *FULL NEUTRAL* | | | | | | | *15 kV 133% INSULATION LEVEL* |
| Size AWG or kcmil | Number of Strands | Insulation Thickness | Full Neutral No. of wires | Full Neutral Size AWG | Jacket Thickness | Approximate Outside Diameter | Approximate Net Weight |
| | | Mils | Mils | Mils | Mils | Inches | Lbs/Mft |
| 2 | 7 | 220 | 10 | 14 | 50 | 1.09 | 547 |
| 1 | 19 | 220 | 13 | 14 | 50 | 1.13 | 620 |
| 1/0 | 19 | 220 | 16 | 14 | 50 | 1.17 | 698 |
| 2/0 | 19 | 220 | 13 | 12 | 50 | 1.25 | 831 |
| 3/0 | 19 | 220 | 16 | 12 | 50 | 1.30 | 949 |
| 4/0 | 19 | 220 | 13 | 10 | 50 | 1.39 | 1150 |
| 250 | 37 | 220 | 16 | 10 | 50 | 1.48 | 1337 |
| 350 | 37 | 220 | 16 | 9 | 50 | 1.60 | 1647 |

The above data are approximate and subject to normal manufacturing tolerances.

| 5 kV - 15 kV 100% or 133% Insulation Level Ampacity | | |
|---|---|---|
| Size AWG or kcmil | † Ampacity | |
| | Full Neutral | 1/3 Neutral |
| 2 | 125 | 125 |
| 1 | 145 | 146 |
| 1/0 | 165 | 166 |
| 2/0 | 188 | 189 |
| 3/0 | 213 | 214 |
| 4/0 | 241 | 244 |
| 250 | - | 266 |
| 350 | - | 320 |
| 500 | - | 394 |
| 750 | - | 476 |
| 1000 | - | 550 |

| 25 kV - 46 kV 100% or 133% Insulation Level Ampacity | | |
|---|---|---|
| Size AWG or kcmil | † Ampacity | |
| | Full Neutral | 1/3 Neutral |
| 1 | 147 | 147 |
| 1/0 | 167 | 167 |
| 2/0 | 193 | 194 |
| 3/0 | 218 | 220 |
| 4/0 | 247 | 250 |
| 250 | - | 273 |
| 350 | - | 326 |
| 500 | - | 392 |
| 750 | - | 486 |
| 1000 | - | 556 |

† Ampacities are based on the following:
Three shielded single insulated aluminum conductors in buried ducts, single circuit, 25°C ambient earth temperature, 75% load factor, 90 RHO and 90°C conductor temperature (IEEE Std. 835).

| Rating Voltage | Minimum Size AWG |
|---|---|
| 8 kV | 6 |
| 15 kV | 2 |
| 25 kV | 1 |
| 35 kV | 1/0 |